Azure status history

This page contains Post Incident Reviews (PIRs) of previous service issues, each retained for 5 years. From November 20, 2019, this included PIRs for all issues about which we communicated publicly. From June 1, 2022, this includes PIRs for broad issues as described in our documentation.

Product:

All

Region:

All

Date:

All

October 2021

RCA - Virtual Machines

Tracking ID: 0NC_-L9G

Summary of Impact: Between 06:27 UTC and 12:42 UTC on 13 Oct 2021, a subset of customers using Windows-based Virtual Machines (Windows VM) may have received failure notifications when performing service management operations - such as start, create, update, delete. Deployments of new VMs and any updates to extensions may have failed. Management operations on Availability Set, Virtual Machine Scale Set were also impacted.

Non-Windows Virtual Machines were unaffected, however services with dependencies on Windows VMs may have also experienced similar failures when creating resources.

Root Cause: Windows-based Virtual Machines utilize the Windows Virtual Machine Agent (VM Agent) extension, which is used to manage interactions between the Virtual Machine and the Azure Fabric.

When creating and updating Windows VMs, the Compute Resource Provider (CRP) has a dependency upon the Platform Image Repository to retrieve download locations for the latest version of the VM Agent package. Using this information, the VM Agent will update itself to the latest version in the VM.

As part of the journey to move all classic resources to Azure Resource Manager (ARM), we are migrating the image and extension publishers to the regional ARM publishing pipeline. Approximately 20% of all extensions have been successfully migrated.

At approximately 06:27 UTC, tooling provided an ARM template for use in performing these migrations. This tooling did not consider an edge case and as an unintended consequence marked the Windows VM Agent extension as visible to the publishing subscription only in the ARM regional service after migration. As the result, VM management operations started to fail after receiving zero results from the regional Platform Image Repositories.

The outcome of this was that service management operations (start, stop, create, delete, etc.) on customers Windows VM were unable to locate the Windows VMAgent extension, and thus unable to complete successfully.

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Part of our change management process is to leverage the Safe Deployment Practice (SDP) framework (https://azure.microsoft.com/en-us/blog/advancing-safe-deployment-practices/). In this case, some of the functionality of our classic infrastructure is incompatible with the SDP framework. This incompatibility underscores the importance in which we are treating the complete migration to ARM. Once the migration is complete, it will allow us to make all changes using the SDP framework without using bespoke tools that support classic resources only.

Mitigation: Determining the root cause took an extended period due to multiple releases for Azure components being in flight simultaneously on the platform, each of which had to be investigated. Additionally, involving subject matter experts (SMEs) for each of the involved components added to this time as we needed to eliminate multiple possible scenarios to ensure we could triage the underlying cause.

Once we determined the issue, and reviewed multiple mitigation options, we mitigated impact by making the extension public in one region at first and validating the results, ensuring no further impact would be caused by a surge in requests for Virtual Machines. Once validated, we started rolling out the change to the new pipeline region-by-region, mitigating the issue. Engineers monitored the platform success rate for operations after the changes were completed.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- The migration of remaining packages in this category (including the Linux version of the VM Agent) is on hold until all repairs are in place
- Additional pre-check and post-checks are being developed and implemented
- VM operation resilience to failures when VM agent cannot be found
- Engineering is also evaluating other safeguards to flight each extension type and prevent any potential negative impact with the remainder of migration.

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September 2021

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Microsoft Azure Portal - Issues while trying to create an application - Mitigated

Tracking ID: 4M8X-VTZ

Summary of Impact: Between 15:00 UTC on 03 Sep 2021 and 01:24 UTC on 09 Sep 2021, customers may have experienced issues while trying to create an application on the Azure portal when signed-in with their Microsoft Account (MSA). This issue had no impact on users who have Azure AD tenants.

Preliminary Root Cause: We determined that this issue was caused due to insufficient capacity to handle requests.

Mitigation: We scaled up the capacity to mitigate the issue.

Next Steps: We will continue to investigate to establish the full root cause and prevent future occurrences. Stay informed about Azure service issues by creating custom service health alerts: https://aka.ms/ash-videos for video tutorials and https://aka.ms/ash-alerts for how-to documentation.

August 2021

RCA - Connection errors for resources leveraging Azure Front Door and Azure CDN

Tracking ID: 0MQY-NPG

Summary of Impact: Between 06:30 UTC and 09:30 UTC on 11 Aug 2021, a subset of customers leveraging Azure Front Door and Azure CDN Standard from Microsoft in Japan East, Japan West, Korea South, Korea Central and/or West US regions may have experienced intermittent HTTPS request connectivity failures when trying to reach their applications. During the incident, the average global error rate was ~2.5% and the peak global error rate was ~5%.

Root Cause: Azure Front Door and Azure CDN Standard from Microsoft serve traffic through edge locations around the world. We were in the process of rolling out a software update to prevent the use of TLS session resumption keys which were older than specific thresholds. The update followed the Azure safe deployment process and was rolling out in phases until it reached the impacted locations. A subset of edge locations in Korea, Japan, and West US were running with stale TLS resumption keys, and the rolled-out update triggered the

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mechanism to prevent the reuse of stale keys. However, a code defect in the rolled-out version resulted in a race condition where a few servers in the impacted locations tried to revert to a full TLS handshake. This race condition resulted in these servers dropping HTTPS requests.

Mitigation: Our monitoring detected this issue and alerted the service team. To mitigate, we removed unhealthy edge locations from serving traffic, which routed traffic to healthy edge locations. We also rolled back the update that caused the regression.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Deploy the fix for the regression that caused the race condition.
- Enhancements to monitoring to ensure alerting if TLS resumption is off or if the session resumption key is older than threshold.
- Enhancements in staging environments to account for additional stale TLS resumption failure modes.

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July 2021

RCA - Azure Network Infrastructure service availability issues - Brazil Southeast

Tracking ID: LNZM-TZG

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Summary of Impact: Between 13:48 UTC and 15:20 UTC on 27 July 2021, a subset of customers experienced issues connecting to their services in the Brazil Southeast region.

Root Cause: We determined that a degradation in connectivity was caused by packet loss when the metadata on one of our regional-level routers was updated incorrectly. As part of a planned network configuration refresh, an update was being performed on the regional-level routers in the Brazil Southeast region. The regional-level tier of the network is designed with redundancy to allow a subset of the routers at that network tier to be taken off-line (not serving customer traffic) for updates.

During the update, our automated network configuration system applied an incorrect IPv4 network prefix (IP Range) to a regional-level router that was taken off-line. Restoring traffic to this regional-level router resulted in packet loss for some of the Azure services in the region. The incorrect network prefix caused traffic from this region to Azure destinations in other regions and a subset of internet regions to be dropped.

Mitigation: The device with incorrect prefixes was removed from service. This mitigation took longer than expected because automated safety checks were failing for the entire region, and some human intervention was required to proceed with the traffic rollback.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- We have audited all ongoing deployments to identify this error pattern and enhancing our validation checks to prevent such a combination of errors, including improved detection logic.
- We are continuously enhancing our alert logic to help identify issues faster and force rollbacks without human intervention.

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RCA - Service availability issue - Brazil South

Tracking ID: 5TGJ-1SZ

Summary of impact: Between 19:00 UTC and 21:28 UTC on 23 Jul 2021, a subset of customers in Brazil South may have experienced issues connecting to a number of Azure services hosted in this region, including, but not limited to, App Service, ExpressRoute, Azure Search, and VPN Gateway.

Root cause: On 23 Jul 2021, at 19:00 UTC, we received an alert from our health monitoring that there were network connectivity issues in the Brazil South region. On initial investigation, we identified a problem in the physical network infrastructure. Upon further investigation, we found that the physical network links to some of the devices within a single cluster were disrupted during a planned maintenance activity. During this maintenance activity, a technician was executing a change that impacted the incorrect set of cables. The work that was executed on site led to the disruption of services.

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Mitigation: The Data Center Operations team worked at restoring some of the physical links which restored network connectivity to operational threshold by 21:20 UTC. By 21:28 UTC most of the impacted services were recovered.

Data Center Operations team then continued to restore the remaining subset of cables and completed this activity at 00:52 UTC on 24 Jul 2021, to bring the services back through normal connectivity.

Next steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Reviewing existing network datacenter operations cabling processes and standards.
- Revisiting and retraining our data center staff on processes.

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RCA - Azure Cognitive Services

Tracking ID: 4LRL-V8G

Summary of impact: Between approximately 07:00 UTC and 13:55 UTC on 21 Jul 2021, customers using Cognitive Services may have experienced difficulties connecting to resources. Impacted customers may have experienced timeouts or 401 authorization errors.

Root Cause: We determined that a subset of cognitive services backend compute clusters were impacted during this incident, causing them to be unable to service customer requests. Our investigation revealed that this issue was triggered by a pending (deployed, but awaiting reboot) OS security update which had introduced a networking change on the updated clusters. The networking update unintentionally caused a networking state which was not supported, and thus when the update was staged, the machines lost network connectivity and could no longer handle requests.

This issue had two separate impact streams as there was a loss of available compute clusters due to the update, and also there was impact to the control plane functionality, thus preventing the dynamic addition of resources to meet with demand. In addition, the standard auto-mitigation workstream of rebooting impacted nodes could not execute, as the nodes could not be contacted due to the networking connectivity issues on the individual nodes. The ultimate outcome was a significant impact to our ability to service Cognitive Services requests for some customers due to loss of compute nodes, and impairment of our ability to auto-mitigate using the standard frameworks.

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Mitigation: Our internal monitoring detected the problem and alerted the on-call engineers. We initially mitigated customer impact by recycling the impacted nodes of the control plane which restored the self-healing capabilities of the system. This mitigation had to be applied on multiple parts of the service as the issue affected several internal subsystems (e.g., authentication, speech-to-text, post-processing, text-to-speech). We then validated that we were able to service customer requests.

Separately, an updated security patch was produced that reverted the unintentional change in behavior such that future critical security patches for this component will not trigger this failure pattern again.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Improvement of the validation mechanisms of update testing to ensure similar issues are detected pre-deployment
- Improvement of the patch staging/pre-reboot process to reduce the footprint of potential impact during updates
- Improvement of the cluster network connectivity detection and automatic remediation/reset processes

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RCA - Azure Cognitive Services - Azure Government

Tracking ID: 4VWK-LPZ

Summary of impact: Between approximately 03:00 EDT and 09:55 EDT on 21 Jul 2021, customers using Cognitive Services may have experienced difficulties connecting to resources. Impacted customers may have experienced timeouts or 401 authorization errors.

Root Cause: We determined that a subset of cognitive services backend compute clusters were impacted during this incident, causing them to be unable to service customer requests. Our investigation revealed that this issue was triggered by a pending (deployed, but awaiting reboot) OS security update which had introduced a networking change on the updated clusters. The networking update unintentionally caused a networking state which was not supported, and thus when the update was staged, the machines lost network connectivity and could no longer handle requests.

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- Improvement of the patch staging/pre-reboot process to reduce the footprint of potential impact during updates
- Improvement of the cluster network connectivity detection and automatic remediation/reset processes

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RCA - Azure Front Door and Azure CDN - Connectivity issues and increased latency

Tracking ID: 0TYG-DPG

Summary of impact: Between approximately 14:55 UTC and 19:50 UTC on July 12th, 2021, a subset of customers primarily in the US and Canada experienced increased latency and connection timeouts when connecting to Azure Front Door and Azure CDN Standard from Microsoft. Retries may have been successful on the resources/environments that experienced issues. The peak failure rate was approximately 70% in some East US metro locations. Aggregate availability drop average across the US and Canada was around 10%. A subset of customers using Windows Virtual Desktop may have experienced connection failures when attempting to access remote applications and resources.

Root Cause: Azure Front Door and Azure CDN Standard from Microsoft service run a periodic background task to process customer configuration updates. We made a recent software update in the background process following our safe deployment guidelines. The release started on July 2nd, 2021 and was completed on Sunday morning July 11th, 2021. A software regression in this release caused intermittent CPU consumption to spike to 40-60% of CPU capacity, approximately 4 times an hour, lasting less than a minute each. This reduced the overall available capacity for processing incoming requests. During the rollout, we did not observe customer traffic availability degradation which allowed the rollouts to proceed. This was because the initial rollout was in edge locations with low traffic and overall reduced traffic volume during the July 4th week in the USA. Peak traffic increase in the busiest metropolitan areas on Monday morning in the USA initiated the incident due to the reduced capacity caused by regression. This resulted in intermittent increased latency, timeouts, and connection failures observed by customers. The Windows Virtual Desktop infrastructure services makes use of Azure Front Door. The increased latency and timeouts mentioned above caused some user requests for these services to fail.

Mitigation: Our internal monitoring detected the drop in availability and issued service alerts. Resource Health Check (RHC) system also detected the drop in availability and issued automated alerts to customers. Our first mitigation was to re-balance incoming traffic to other nearby edge locations to alleviate the issue. However, we observed that even though re-balancing traffic reduced the incident's severity, the overall CPU consumption on some edge sites remained high. We then further diagnosed the issue to be related to the recent background service update. A second mitigation was then applied to limit CPU utilization on the background service to ensure that it does not go over a threshold. The combination of re-balancing traffic and reducing CPU consumption limit for background service successfully mitigated the incident. We temporarily reconfigured the Windows Virtual Desktop infrastructure services to bypass the use of Azure Front Door during a portion of the outage period.

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Next Steps: We sincerely apologize for this incident and any inconvenience it may have caused to our customers. In our constant effort to improve the stability of the Azure platform, we have undertaken the following repair items:

- Completed changes to enforce resource utilization limits to help prevent impact to capacity.
- Apply fix to regression that caused high CPU utilization.
- Improve monitoring/alerting to help improve detection of resource usage anomalies.
- Apply additional resource limits on applicable background services to help prevent capacity impact to other services.
- Fine tune resource health alerts to include a list of affected edge locations.
- Enhancements to traffic management to better redirect incoming requests from edge locations that have high CPU.

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RCA - Service management operation failures - North Europe

Tracking ID: 0_JL-9SG

Summary of impact: Between 21:19 UTC on 07 Jul 2021 and 12:10 UTC on 08 Jul 2021, a subset of customers in North Europe may have intermittently received errors when performing service management operations for services that rely on compute resources in this region.

Root Cause:

The request queue limit on some frontends for the disk management that is responsible for service management operations in North Europe reached their limits and were rejecting a subset of requests to it.

This resulted in intermittent failures for service management requests for virtual machines and disks. The failed requests generally succeeded on retries. The issue was triggered by a platform update on the servers running the disk management service. The platform update caused expected batched failovers between the replicas of the disk management service which unexpectedly led to high latency for some calls and build-up of queues on the service front-ends. This in turn resulted in higher latency and failures for subsequent calls. The detailed sequence of events which led to this situation is under investigation.

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Mitigation:

Automated service monitoring raised immediate alerts for customer impacting failures and engineers for the impacted services were engaged immediately. This was a complex situation where a number of events and changes needed to be investigated by engineers from multiple Azure components. Throughout the incident, multiple mitigations strategies were deployed with partial success. Finally, the

mitigation to perform an upgrade of service frontend components fully mitigated the issue. It was once completed the frontends were re-initialized, that the issue was fully mitigated.

Next Steps

- We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case we are adding instrumentation to our service code to allow quicker diagnosis and mitigation for similar issues.

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June 2021

RCA - Azure Resource Manager - Degraded Performance managing resources

Tracking ID: 1V9K-PSZ

Summary of Impact: Between as early as 22:24 UTC on 29 Jun 2021 and 14:30 UTC on 30 Jun 2021, a subset of Azure customers may have experienced intermittent errors when attempting to access the Azure portal and other Microsoft and Azure services. Impact was observed across multiple services and regions to varying degrees. Recovery time varied by service, and most services fully recovered by 14:30 UTC. During this time, retries may have succeeded.

Root Cause: Azure Resource Manager service has a configuration to connect to a required backend storage. The connection to some of these backend storage endpoints started failing after a maintenance to update their configuration was deployed. Over time, these machines naturally restarted as needed, picking up the new configuration - which contained an issue. Once restarted, these machines would fail to connect to the storage endpoints. Over time, this led to degraded service performance in the regions where this configuration was rolled out to, as a subset of machines would become affected.

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The configuration change was designed to be non-impacting and thus was rolled out in parallel, leading to impact across more than one region.

Mitigation: Once we detected the issue, we stopped the rollout of the new configuration. In order to mitigate the issue, we took unhealthy nodes out of rotation, and patched and rolled out the correct configuration in a staged manner. After mitigation at 14:30 UTC, we continued to monitor the platform to ensure stability of the service after the rollout.

Next Steps: We sincerely apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Review of requirements regarding safe deployment for configuration changes
- Updates to prevent this component from failing the rest of the service in the event of an issue
- Adding this update scenario to automation to make configuration changes when required

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English (UK)

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